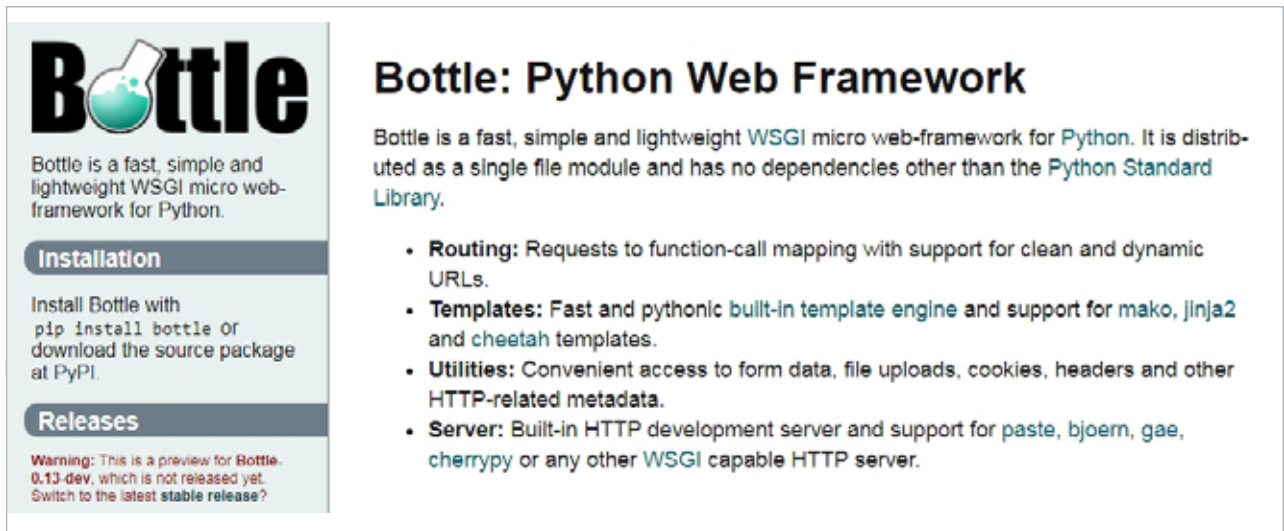




Bottle: Python Web Framework

Bottle is a fast, simple and lightweight WSGI micro web-framework for Python. It is distributed as a single file module and has no dependencies other than the Python Standard Library.

- **Routing:** Requests to function-call mapping with support for clean and dynamic URLs.
- **Templates:** Fast and pythonic built-in template engine and support for mako, Jinja2 and cheetah templates.
- **Utilities:** Convenient access to form data, file uploads, cookies, headers and other HTTP-related metadata.
- **Server:** Built-in HTTP development server and support for paste, bjoern, gae, cherrypy or any other WSGI capable HTTP server.

Figure 2: Official portal of Bottle framework

Flask framework

Flask is a lightweight but powerful framework for the development of Web applications using the Python programming environment. Complex and high performance Web applications can be developed and deployed using Flask very effectively. Very few lines of code are needed to program real-world applications, and there are no special requirements of dependencies on other libraries.

To work with the Flask framework in Python, the following instruction is executed first in a command prompt:

```
$ pip install flask
```

After the integration of Flask with the existing Python environment, the classical Web application in Python Flask is written in the following format:

```
from flask import Flask, escape, request
app = Flask(__name__)
@app.route('/')
def myfn():
    myvar = request.args.get("myvar",
"World")
    return f'Hello, {escape(myvar)}!'
```

The script is executed in a command prompt using the following:

```
$ flask run
```

The execution can be viewed on a Web browser with the URL *http://127.0.0.1:5000*.

The performance of the Flask framework is excellent for the object relational model (ORM) and the structured query language that are used for real-time applications. In addition, the execution time of Flask is very less.

Bottle framework

Bottle is another backend Web framework that is used for browser based applications. Simplicity and a single file distribution process are the key advantages of Bottle. In addition, it offers a lot of flexibility to expand its features, including many real-time features like templating and routing.

To install the Bottle framework in the Python environment, the following instruction is executed:

```
$ pip install bottle
```

The following is a programming paradigm in the Bottle framework that is quite simple and easy to execute:

```
from bottle import run, route, template
@route('/hello/<myvar>')
def index(myvar):
    return template('<b>Hello
{{myvar}}</b>!', myvar=myvar)
run(host='localhost', port=8080)
```

This code is executed on port 8080 on localhost or on URL of 127.0.0.1 which can be customised as per the requirements. If the script needs to be executed on the cloud, the IP address is modified.

There is tremendous scope for using these Python frameworks in Web application development. The inclusion of database engines in these frameworks ensures that records can be managed effectively. 

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